

Data Visualization Workshop

Practical Statistics for Business

Learning Objectives

1. Understand visualization types
2. Interpret data charts
3. Create effective visualizations
4. Communicate insights

Chart Types

Chart Type	Use Case
Bar Chart	Comparing categories
Line Chart	Trends over time
Scatter Plot	Relationships between vars
Boxplot	Distribution comparison

Variance Analysis

Boxplot Components

- Q1, Q2 (median), Q3
- Whiskers: $1.5 \times \text{IQR}$
- Outliers: Beyond whiskers

Efficiency Metrics

Discussion

Different visualization methods reveal different insights.

Success Factors

Exercise

How does debt affect subscription likelihood?

Work in small groups for 10 minutes.

Income Distribution

Best Practices

- ✓ Always visualize data first
- ✓ Choose appropriate chart type
- ✓ Label axes and provide context
- ✓ Use color purposefully
- ✓ Tell a story with data

Further Reading

- *Think Stats* by Allen B. Downey
- *Data Science for Business* by Provost
- *Python for Data Analysis* by Wes McKinney

Questions & Discussion

Thank you for attending!

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